JOHANNES KEPLER UNIVERSITY LINZ

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Assignment 1

Random Variables and Vectors

This problem sheet has to be solved individually from the teams. In case plagiarism is encountered there will be no points granted for all involved teams, no exclusions. Document your results (derivations, diagrams, answers, ...) carefully in a **single** protocol (see *course_info.pdf*). Submission deadline (Moodle upload): **Thursday, November 19 2026, 08:00**.

Exercise 1 *Gradient of quadratic form* (8 P) *A1P13*

Let

$$f(\boldsymbol{x}) = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x},$$

where $\boldsymbol{x}$ is a vector of length N and $\boldsymbol{A}$ is a *symmetric* $N \times N$ matrix. Show that

$$\frac{\partial f}{\partial \boldsymbol{x}} = 2\boldsymbol{A}\boldsymbol{x}.$$

(a) Rewrite $f(\boldsymbol{x})$ such that there are no more vectors used, i.e. use two sums to perform the matrix/vector multiplication

(b) Compute the gradient $\frac{\partial f}{\partial \boldsymbol{x}}$. Explicitly write each step of derivation.
Hints:

- Use the product rule for computation of the gradient
- As $\boldsymbol{A}$ is a *symmetric* matrix $\boldsymbol{A} = \boldsymbol{A}^T$ or $A_{ij} = A_{ji}$

Exercise 2 *The Variance* (6 P) *A1P17*

Prove that $Var(X) = E(X^2) - E(X)^2$.

Exercise 3 *Auto-covariance matrix and correlation coefficient of 2-dimensional PMF* (10 P) *A1P16*

Consider the joint PMF of a random vector $\mathbf{X} = [X_1, X_2]^T$ given in the Table 1 below.

	$x_2 = 0$	$x_2 = 1$	$x_2 = 2$
$x_1 = 0$	$\frac{1}{8}$	0	$\frac{1}{4}$
$x_1 = 1$	0	$\frac{1}{8}$	$\frac{1}{4}$
$x_1 = 2$	$\frac{1}{8}$	0	$\frac{1}{8}$

Table 1: Joint PMF $p_{\mathbf{X}}[x_1, x_2]$

(a) Sketch the marginal PMFs $p_{X_1}[x_1]$ and $p_{X_2}[x_2]$

(b) Sketch the conditional PMF $p_{X_2|X_1}[x_2|x_1 = 1]$

(c) Determine the mean vector $\boldsymbol{\mu}_{\mathbf{X}}$

(d) Determine the auto-covariance matrix $\mathbf{C}_{\mathbf{XX}}$

(e) Determine the correlation coefficient between X_1 and X_2

Exercise 4 *Covariance and correlation coefficient of linearly dependent random variables* (8 P) *A1P08*

Analytically derive the covariance and correlation coefficient between X and Y, given

$$Y = aX + b,$$

where a, b are constants and X is a random variable with $E(X) = 0$. Also, compute the correlation coefficient for $a < 0$ and $a > 0$.

Note 1 *Correlation between random variables does not imply a causal relationship between the random variables!* *A1Note01*

Source: Kay, Intuitive Probability and Random Processes, Springer, p. 197
A frequent misapplication of probability is to assert that two quantities that are correlated $\rho_{X,Y} \neq 0$ are such because one causes the other. To dispel this myth consider a survey in which all individuals older than 55 years of age in the U.S. are asked whether they have ever had prostate cancer and also their height in inches. Then, for each height in inches we compute the average number of individuals per 1000 who ever had cancer. If we plot the average number, also called the incidence of cancer, versus height, a typical result would be as shown in Fig. 1. This indicates a strong positive correlation of cancer with height. One might be tempted to conclude that growing taller causes prostate cancer. This is of course nonsense. What is actually shown is that segments of the population who are tall are associated with a higher incidence of cancer. This is because the portion of individuals who are taller than the rest are predominately male. Females are not subject to prostate cancer, as they have no prostates! In summary, correlation between two variables only indicates an *association*, i.e., if one increases, then so does the other (if positively correlated). No physical or causal relationship need exist.

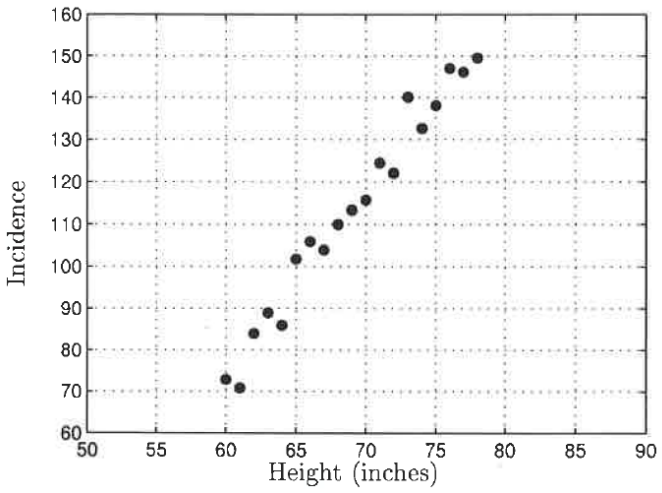


Fig. 1: Incidence of prostate cancer per 1000 individuals older than age 55 versus height.

Exercise 5 *Correlation vs Causality* (8 P) *A1P24*

Correlation is a measure of linear dependency, and it is often falsely implied as causality. A popular example for this is that it was assumed that the child birth rate is related to the number of storks in a country (refer to Table 2[1]).

Country	Area [km^2]	Storks	Inhabitants [10^6]	Child birth rate (10^3/Year)
Austria	83860	300	7.6	87
Poland	312680	30000	38	610
Switzerland	41290	150	6.7	82
Hungary	93000	5000	11	124
Spain	504750	8000	39	439
Greece	132000	2500	10	106
Portugal	92390	1500	10	120
Denmark	43100	9	5.1	59

Table 2: Storks and child birth rates for different countries.

(a) Plot the child birth rate over the number of storks

(b) Compute the correlation coefficient between the child birth rate and the number of storks. Do you think there is a causal relationship or is it spurious correlation (*Scheinkorrelation*)?

(c) Now compute the correlation coefficient between the area of the country and the number of storks and compare it to the result from (b). What do you observe?

[1]Data taken from http://www.math.uni-paderborn.de/agbiehler/sis/sisonline/struktur/jahrgang21-2001/heft2/Langfassungen/2001-2_Matth.pdf